

Challenge 4 · CoGNETs · Smart Edge Resource Auctions

Swarm Lab

Compete today. Stay alive tomorrow.

A battery-aware edge agent

Los Bocatones · Marc and Carlos

GitHub repo

<https://github.com/carlosclariana/veleshack-2026-4th-challenge>

TEAM_NAME	los-bocatones
Dockerfile	agent-template/Dockerfile
Members	Marc and Carlos
Fork of	czavitsanos-iti/ veleshack-2026-4th-challenge

Build and run

```
docker build -t swarm-agent:submission \
    agent-template
```

```
docker compose --profile agent up --build
```

Live dashboard at <http://localhost:8080>

8 / 8 official conformance checks on the built image

Summary

The energy you win is drawn from your own battery. Run flat and you sit the round out. Our agent treats charge as money and bids for the whole match, not for one round.

Reads the market

Rebuilds rival bids from the shares it was actually allocated.

Prices the recharge

Scores each candidate bid by utility per active and recharge round.

Protects the floors

Missing a service floor halves the round, so candidates sit on the safe side.

Rounds spent resting, out of 60

Starting template

18.4 resting · 13.39 points



Our agent

6.5 resting · 17.23 points



■ resting on a flat battery · mean of 50 held-out seeds

Highlights

+28.7 %

Over the starting template

17.23 against 13.39 mean score on 50 held-out seeds.

50 / 50

Ahead of all three bots

In every held-out run. Policy frozen after 15 development seeds.

1st of 4

Live graded match over HTTP

18.24 against 13.51 for the best bot. 87 injected faults, no round missed.

What did not work

Saving more battery

Charging 1.5× and 2× more for energy scored lower. Rejected.

First live run: 2nd

History arrived incomplete. We fixed the transport and kept the result.

Neighbours lose

Their combined score fell from 44.24 to 38.41.